

Differential Calculus — Chain Rule

Chapter 2: Radical Function of Cubic Polynomial | Page 33

PROBLEM STATEMENT

Square Root & Cubic Composite

Find the derivative with respect to x of the function:

$$y = \sqrt{2x^3 - 1}$$

STEP 1 প্রদত্ত অমূলদ ঘাত ফাংশনটি উল্লেখ করি (Given Radical Function)

$$y = \sqrt{2x^3 - 1} = (2x^3 - 1)^{1/2}$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [\sqrt{2x^3 - 1}]$$

STEP 3 বর্গমূলীয় চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[\sqrt{u}] = \frac{1}{2\sqrt{u}} \frac{du}{dx}$)

$$\frac{dy}{dx} = \frac{1}{2\sqrt{2x^3 - 1}} \cdot \frac{d}{dx}(2x^3 - 1)$$

STEP 4 ভেতরের ত্রিঘাত অংশের অন্তরক নির্ণয় করি (Differentiating Inner Cubic Expression)

$$\frac{d}{dx}(2x^3 - 1) = 2(3x^2) - 0 = 6x^2$$

STEP 5 মান বসিয়ে লব ও হরে ২ কাটাকাটি করে সরলীকরণ করি (Simplification)

$$\frac{dy}{dx} = \frac{1}{2\sqrt{2x^3 - 1}} \times 6x^2 = \frac{3x^2}{\sqrt{2x^3 - 1}}$$

FINAL ANSWER
$$\frac{dy}{dx} = \frac{3x^2}{\sqrt{2x^3 - 1}}$$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 03

BOX 1: FORMULATION & RADICAL DIFFERENTIATION RULES

- **Direct Radical Formula:** $\frac{d}{dx} \sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}$.
- **Power Rule Equivalence:** Writing $y = [f(x)]^{1/2} \implies y' = \frac{1}{2} [f(x)]^{-1/2} \cdot f'(x) = \frac{f'(x)}{2\sqrt{f(x)}}$.
- For $f(x) = 2x^3 - 1$, $f'(x) = 6x^2 \implies \frac{dy}{dx} = \frac{6x^2}{2\sqrt{2x^3-1}} = \frac{3x^2}{\sqrt{2x^3-1}}$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Canceling Factor Error:** Forgetting to divide 6 by 2, resulting in an erroneous $6x^2/\sqrt{2x^3-1}$.
- **Missing Power on x:** Differentiating $2x^3$ to $6x$ instead of $6x^2$. Always double-check $\frac{d}{dx}(x^3) = 3x^2$.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Real Domain:** For the square root to exist in $\mathbb{R}$, we need:

$$2x^3 - 1 \geq 0 \implies x^3 \geq \frac{1}{2} \implies x \geq \frac{1}{\sqrt[3]{2}} \approx 0.7937$$

Hence, $\text{Domain}(y) = [\frac{1}{\sqrt[3]{2}}, \infty)$.

- **Differentiability Domain:** Since $\sqrt{2x^3-1}$ is in the denominator of $\frac{dy}{dx}$, the function is strictly differentiable on $(\frac{1}{\sqrt[3]{2}}, \infty)$.
- **Vertical Tangent:** As $x \rightarrow (\frac{1}{\sqrt[3]{2}})^+$, $\frac{dy}{dx} \rightarrow +\infty$, indicating a vertical tangent at the boundary point.

BOX 4: LOGARITHMIC DIFFERENTIATION METHOD (ALTERNATIVE)

- Given $y = (2x^3 - 1)^{1/2} \implies \ln y = \frac{1}{2} \ln(2x^3 - 1)$.
- Differentiating both sides w.r.t. x :

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{2} \cdot \frac{6x^2}{2x^3 - 1} = \frac{3x^2}{2x^3 - 1}$$

- Multiplying by $y = \sqrt{2x^3 - 1}$:

$$\frac{dy}{dx} = \sqrt{2x^3 - 1} \cdot \frac{3x^2}{2x^3 - 1} = \frac{3x^2}{\sqrt{2x^3 - 1}}$$

BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct setup using $\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}} \frac{du}{dx}$.
- **1.0 Mark:** Accurate computation of $\frac{d}{dx}(2x^3 - 1) = 6x^2$.
- **1.0 Mark:** Correct reduction and final boxed fraction form $\frac{3x^2}{\sqrt{2x^3-1}}$.